

# **Oxidative Stress and Antioxidant Defence**

Antioxidant enzymes (SOD, catalase, glutathione peroxidases), synthesis of glutathione (GCLC, GCLM), NRF2 and FOXO3, phase detoxification II (NQO1, GST), paraoxonase, heme-oxygenase and sources of oxidants (NOS3, MPO, CYBA)

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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# 1. Summary

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**What was seen.** 31 SNPs were read in 19 genes (23 chip, 7 imputed, 1 unread) and 5 polygenic scores. The portrait is of master regulators and strong main enzymes, with a phase detoxification II and a weaker paraoxonase.

**Strong points.** Homozygous FOXO3 for the longevity allele in the main 4 SNPs (and 1 favourable allele in rs12212067); **NRF2 –617C/C** (without the allele that stops self-induction); **SOD2 Ala/Ala** (MnSOD more active); **GCLC –129C/C** (glutathione synthesis without the weak allele); **NQO1 \*1/\*1** (full activity); GSTP1 without Val114; **HMOX1 –413A/A**; APOE without ε4. The longevity score (score A) is in p92.4.

**Weak points.** Probable null GSTM1 (A/A deletion marker) and **GSTA1 \*B/\*B** (~3-4x minus GSTA1 in the liver): the conjugation of epoxides and lipid peroxidation products depends more on GSTP1 and glutathione. PON1 with low expression (–108T/T and Met55/Met55), partially compensated by Gln192Gln. **GPX1 Pro198Leu** and **GCLM –588T** heterozygous; **NRF2 –653G** (1 copy) and –651A (called the uncertain chip). **MPO –463G/G** (more myeloperoxidase) and NOS3 –786C and Asp298 heterozygous (less nitric oxide).

**Reading.** The SOD2 Ala/Ala forms H<sub>2</sub>O<sub>2</sub> faster and the GPX1 Leu198 removes it a little slower: the equilibrium depends on catalase (typical), glutathione and selenium. Null GSTM1 and GSTA1 \*B/\*B make exposure to smoke, burned meat and pollutants more relevant than to the average.

## 2. How to read this report

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- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality  $DR2 \geq 0.8$ . Under this threshold, or outside the imputed zone, SNP is treated as SNP not found in the data.
- **Variants:** 31 SNPs: 23 of the chip, 7 imputed and 1 not found. All genotypes were reread directly in the raw DNA and in the imputation for this report.
- **Strand:** All genotypes are in the GRCh38 strand +. 0 SNPs needed conversion to complementary strand and 2 are palindromic (A/T or C/G). When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Chromosome X:** Sérgio F has only one copy of X outside the pseudoautosomal regions, so the SNPs of X appear with a single letter (hemizygous).
- **Polygenic scores (PGS):** 5 PGS Catalog scores calculated locally with 120,035 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ( $n = 633$ ), calculated with exactly the same variants. Percentage  $\geq 80$ : MORE LIKELY/HIGH;  $\leq 20$ : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

### 3. Polygenic scores

Percentages in the reference population. The bilirubin is an endogenous antioxidant (Gilbert's syndrome, UGT1A1\*28 heterozygous, is already known). The longevity score B has only 4 variants.

#### Endogenous antioxidants and longevity

| Trait                           | Score (PGS Catalog)              | Variants used                            | Percentile | Result      | Judgement        |
|---------------------------------|----------------------------------|------------------------------------------|------------|-------------|------------------|
| Total bilirubin (score A)       | PGS002160 (Am J Hum Genet 2021)  | 118,548 120,068 (weighted coverage 100%) | 69.5       | TYPICAL     | typical          |
| Total bilirubin (score B)       | PGS000697 (Nat Genet 2021)       | 1,061 1,159 (weighted coverage 96%)      | 71.4       | TYPICAL     | typical          |
| Uric acid                       | PGS000126 (Nat Genet 2019)       | 110 114 (weighted coverage 91%)          | 40.9       | TYPICAL     | without judgment |
| Longevity (score A)             | PGS000906 (J Gerontol A 2021)    | 312 330 (weighted coverage 90%)          | 92.4       | MORE LIKELY | favourable       |
| Longevity (score B, 4 variants) | PGS002795 (Clin Infect Dis 2021) | 4 4 (weighted coverage 100%)             | 47.2       | TYPICAL     | typical          |

## 4. Variants by theme

For each SNP: genotype (strand +), origin, effect allele and number of copies, judgment and evidence. In the "protective" SNPs, having the allele is favourable.

### Primary antioxidant enzymes

SOD2 (mitochondria) and SOD3 (out of cells) transform superoxide into H<sub>2</sub>O<sub>2</sub>; catalase and glutathione peroxidases (GPX1, GPX4) transform H<sub>2</sub>O<sub>2</sub> into water. The balance between the two steps is what counts.

| Gene           | SNP       | Variant                        | Genotype                  | Source                                         | Effect allele (copies) | Judgement            | Allele effect                                                                                                                                         | Evidence |
|----------------|-----------|--------------------------------|---------------------------|------------------------------------------------|------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| SOD2 (MnSOD)   | rs4880    | Val16Ala (Ala = G in strand +) | G/G                       | chip                                           | G (2 copy/s)           | favourable           | Ala16: more efficient mitochondrial import and ~30–40% more active MnSOD (the H <sub>2</sub> O <sub>2</sub> formed must be removed by GPX1/ catalase) | High     |
| SOD3 (EC-SOD)  | rs1799895 | Arg213Gly                      | SNP not found in the data | SNP not found in the data (DR2<0.8 imputation) | G faux copy/s)         | without data         | Gly213 releases EC-SOD from the vascular wall; associated with coronary disease                                                                       | High     |
| CAT (catalase) | rs1001179 | –262C>T                        | C/C                       | chip                                           | T (0 copy/s)           | neutral              | Promoter; discordant direction between studies                                                                                                        | Moderate |
| GPX1           | rs1050450 | Pro198Leu                      | G/A                       | chip                                           | A (1 copy/s)           | unfavourable (light) | Leu198: ~20–40% less activity and lower selenium response                                                                                             | High     |
| GPX4           | rs713041  | c.718T>C (SECIS)               | T/C                       | chip                                           | T (1 copy/s)           | without judgment     | Changes the selenium incorporation priority; heterozygous = intermediate                                                                              | Moderate |

### Glutathione: synthesis and recycling

Glutathione is the main antioxidant within cells. GCL (subunits GCLC and GCLM) is the limiting step of its synthesis; NAC provides cysteine.

| Gene | SNP        | Variant | Genotype | Source | Effect allele (copies) | Judgement            | Allele effect                                                               | Evidence |
|------|------------|---------|----------|--------|------------------------|----------------------|-----------------------------------------------------------------------------|----------|
| GCLC | rs17883901 | –129C>T | G/G      | chip   | A (0 copy/s)           | favourable           | –129T reduces 40–50% the limiting step of glutathione synthesis             | High     |
| GCLM | rs41303970 | –588C>T | G/A      | chip   | A (1 copy/s)           | unfavourable (light) | –588T reduces ~40% the promoter of the modulatory subunit: less glutathione | High     |

### Master regulators (NRF2 and FOXO3)

NRF2 binds more than 200 antioxidant and detoxification genes; FOXO3 induces SOD2, catalase and autophagy and is the most replicated longevity gene.

| Gene          | SNP        | Variant | Genotype | Source              | Effect allele (copies) | Judgement            | Allele effect                                                                   | Evidence |
|---------------|------------|---------|----------|---------------------|------------------------|----------------------|---------------------------------------------------------------------------------|----------|
| NFE2L2 (NRF2) | rs6721961  | –617C>A | G/G      | imputation DR2=1.00 | T (0 copy/s)           | favourable           | –617A reduces the self-induction of NRF2 (the strongest effect of the promoter) | High     |
| NFE2L2 (NRF2) | rs35652124 | –653A>G | T/C      | chip                | C (1 copy/s)           | unfavourable (light) | –653G modestly reduces the promoter                                             | Moderate |

| Gene          | SNP        | Variant  | Genotype | Source | Effect allele (copies) | Judgement    | Allele effect                                                                    | Evidence |
|---------------|------------|----------|----------|--------|------------------------|--------------|----------------------------------------------------------------------------------|----------|
| NFE2L2 (NRF2) | rs6706649  | -651G>A  | T/T      | chip   | T (2 copy/s)           | unfavourable | -651A reduces the promoter (called chip with uncertain quality)                  | Low      |
| FOXO3         | rs2802292  | intronic | G/G      | chip   | G (2 copy/s)           | favourable   | G = more replicated longevity allele; FOXO3 induces SOD2, catalase and autophagy | High     |
| FOXO3         | rs2764264  | intronic | C/C      | chip   | C (2 copy/s)           | favourable   | C = longevity allele                                                             | High     |
| FOXO3         | rs13217795 | intronic | C/C      | chip   | C (2 copy/s)           | favourable   | C = longevity allele                                                             | High     |
| FOXO3         | rs2802288  | intronic | A/A      | chip   | A (2 copy/s)           | favourable   | A = longevity allele                                                             | High     |
| FOXO3         | rs12212067 | intronic | T/G      | chip   | G (1 copy/s)           | favourable   | G: more TGF- $\beta$ 1 and less TNF in monocytes (lighter inflammatory course)   | Moderate |

## II phase detoxification (NQO1 and GST)

NQO1 reduces quinones without radical formation; glutathione S-transferases (GSTP1, GSTA1, GSTM1) combine electrolytes and lipid peroxidation products.

| Gene  | SNP       | Variant         | Genotype | Source | Effect allele (copies) | Judgement        | Allele effect                                                                                                            | Evidence |
|-------|-----------|-----------------|----------|--------|------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------|----------|
| NQO1  | rs1800566 | Pro187Ser (*2)  | G/G      | chip   | A (0 copy/s)           | favourable       | *2: almost zero enzyme (reduces quinones and keeps ubiquinol and vitamin E reduced)                                      | High     |
| NQO1  | rs1131341 | Arg139Trp (*3)  | G/G      | chip   | A (0 copy/s)           | favourable       | *3: splicing altered, less activity                                                                                      | High     |
| GSTP1 | rs1695    | Ile105Val       | A/G      | chip   | G (1 copy/s)           | without judgment | Val105: substrate-dependent activity                                                                                     | High     |
| GSTP1 | rs1138272 | Ala114Val       | C/C      | chip   | T (0 copy/s)           | favourable       | Val114: less activity                                                                                                    | High     |
| GSTA1 | rs3957357 | -69 (*B)        | A/A      | chip   | A (2 copy/s)           | unfavourable     | B/B: ~3-4x minus GSTA1 in liver (GST with peroxidase activity)                                                           | High     |
| GSTM1 | rs366631  | deletion marker | A/A      | chip   | A (2 copy/s)           | unfavourable     | Null GSTM1 marker (without enzyme GSTM1: epoxides, PAH and peroxidation products); does not measure the number of copies | Moderate |

## Paraoxonase and hemeoxygenase

PON1, bound to HDL, protects lipoproteins from oxidation; heme-oxygenase (HMOX1) forms bilirubin and CO, endogenous antioxidants.

| Gene | SNP      | Variant   | Genotype | Source              | Effect allele (copies) | Judgement    | Allele effect                                                                                         | Evidence |
|------|----------|-----------|----------|---------------------|------------------------|--------------|-------------------------------------------------------------------------------------------------------|----------|
| PON1 | rs662    | Gln192Arg | T/T      | chip                | C (0 copy/s)           | mixed        | QQ (without Arg): better protects lipoproteins from peroxidation, but less organophosphate hydrolyses | High     |
| PON1 | rs854560 | Leu55Met  | T/T      | imputation DR2=1.00 | T (2 copy/s)           | unfavourable | Met55: less PON1 in serum and less stable                                                             | High     |
| PON1 | rs705379 | -108C>T   | A/A      | imputation DR2=0.97 | A (2 copy/s)           | unfavourable | -108T: ~2x less expression (main genetic determinant of PON1 levels)                                  | Moderate |

| Gene  | SNP       | Variant | Genotype | Source                 | Effect allele (copies) | Judgement  | Allele effect                                                        | Evidence     |
|-------|-----------|---------|----------|------------------------|------------------------|------------|----------------------------------------------------------------------|--------------|
| HMOX1 | rs2071746 | -413A>T | A/A      | imputation<br>DR2=0.94 | T (0 copy/s)           | favourable | -413A: plus basal HO-1 (bilirubin and CO antioxidants); T gives less | Moderate-low |

## Oxidant sources and nitric oxide

Oxidant sources (neutrophil myeloperoxidase, NADPH-oxidase) and eNOS, which form nitric oxide; when decoupled, it forms superoxide. APOE ε4 as reference for cerebral lipid peroxidation.

| Gene           | SNP       | Variant                     | Genotype | Source                 | Effect allele (copies) | Judgement            | Allele effect                                                             | Evidence |
|----------------|-----------|-----------------------------|----------|------------------------|------------------------|----------------------|---------------------------------------------------------------------------|----------|
| NOS3 (eNOS)    | rs2070744 | -786T>C                     | C/T      | chip                   | C (1 copy/s)           | unfavourable (light) | -786C: ~50% less activity of eNOS promoter and less nitric oxide          | High     |
| NOS3 (eNOS)    | rs1799983 | Glu298Asp                   | T/G      | imputation<br>DR2=1.00 | T (1 copy/s)           | unfavourable (light) | Asp298: eNOS more sensitive to cleavage and decoupling                    | Moderate |
| NOS3 (eNOS)    | rs3918226 | -665C>T                     | C/C      | chip                   | T (0 copy/s)           | favourable           | Allele of promoter associated with endothelial dysfunction in GWAS        | High     |
| MPO            | rs2333227 | -463G>A (G = C in strand +) | C/C      | imputation<br>DR2=1.00 | C (2 copy/s)           | unfavourable         | -463G: more myeloperoxidase (more hypochlorous acid and oxidation of LDL) | High     |
| CYBA (p22phox) | rs4673    | His72Tyr (242C>T)           | G/G      | chip                   | A (0 copy/s)           | neutral              | 242T: less NADPH-oxidase activity in part of the studies                  | Low      |
| APOE           | rs429358  | ε4 (Cys112Arg)              | T/T      | imputation<br>DR2=0.99 | C (0 copy/s)           | favourable           | ε4: lower antioxidant capacity and more cerebral lipid peroxidation       | High     |

## 5. Favourable

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- **FOXO3**: homozygous for the longevity allele in rs2802292 (G/G), rs2764264 (C/C), rs13217795 (C/C) and rs2802288 (A/A); 1 favourable allele in rs12212067.
- **NRF2 –617C/C** (rs6721961 G/G, imputation DR2 1.00): without the allele that reduces self-induction.
- **SOD2 Ala16/Ala16** (rs4880 G/G): MnSOD more active.
- **GCLC –129C/C**: glutathione synthesis without the weak allele.
- **NQO1 \*1/\*1** (without \*2 or \*3) and GSTP1 without Val114.
- **HMOX1 –413A/A** (imputed): more basal heme-oxygenase.
- **NOS3 –665** without the risk allele; APOE without ε4; **catalase –262C/C** (typical).
- **Longevity (score A) p92.4**; bilirubin p69.5–71.4 (typical, to tend to high).

## 6. Unfavourable

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- **Probable null GSTM1** (rs366631 A/A; marker, does not measure copies) and **GSTA1 \*B/\*B** (rs3957357 A/A).
- **Low expression PON1:** –108T/T (rs705379 A/A) and Met55/Met55 (rs854560 T/T), both imputed.
- **GPX1 Pro198Leu** (1 copy): less glutathione peroxidase and less selenium response.
- **GCLM –588T** (1 copy): less subunit modulating GCL.
- **NRF2:** –653G (1 copy) and –651A (T/T; chip call with uncertain quality).
- **MPO –463G/G** (imputed): more myeloperoxidase.
- **NOS3:** –786C (1) and Asp298 (1), less endothelial nitric oxide.

## 7. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

| Mechanism                              | Genetic finding                                                                                                 | Meaning              |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------|
| Oxidative stress                       | SOD2 Ala/Ala, GCLC –129C/C, NQO1 1/1 and NRF2 –617C/C strong; Null GSTM1, GSTA1 B/B, GPX1 Leu198 and lower PON1 | mixed                |
| Redox imbalance                        | Rapid SOD2 + GPX1 Leu198: more transient H2O2; glutathione and selenium decide the balance                      | unfavourable (light) |
| Mitochondrial dysfunction              | MnSOD more active (SOD2 Ala/Ala); NQO1 keeps the ubiquinol reduced                                              | favourable           |
| Autophagy / Deregulated Macroautophagy | Homozygous FOXO3 of longevity (induces autophagy)                                                               | favourable           |
| Nutrient Sensing Deregulation          | FOXO3 (via insulin/IGF-1) with longevity alleles; longevity score p92.4                                         | favourable           |
| Vascular Aging                         | NOS3 –786C and Asp298 heterozygous; MPO –463G/G; Low expression PON1                                            | unfavourable (light) |
| Arterial stiffness                     | Less endothelial nitric oxide (NOS3) and more oxidized LDL (MPO, PON1)                                          | unfavourable (light) |
| DNA Damage and DNA Repair Deficit      | GSTM1 null and GSTA1 B/B: less conjugation of epoxides and PAH (DNA fertilizers)                                | unfavourable         |
| Chronic Inflammation                   | FOXO3 rs12212067 G (less TNF in monocytes); MPO –463G/G (more neutrophil oxidants)                              | mixed                |
| Neuroinflammation                      | APOE without ε4 (less cerebral lipid peroxidation)                                                              | favourable           |

## 8. Practical suggestions

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All suggestions are suggestion (not applied): the real stack of Sérgio F remains as it is, and any change should be discussed with the doctor.

1. **Avoid smoking and burned meat** (carbonized grilles, smoked, tobacco smoke, combustion at home): GSTM1 null and GSTA1 \*B/\*B conjugate worse PAH and epoxides. Cooking at the lowest temperature (stuttering, steam, oven). *Suggestion (not applied).*
2. **Crucifers almost every day** (broccoli, kale, rocket, cress; broccoli shoots): the sulforaphane activates NRF2 (–617C/C responds well) and induces the remaining GST. *Suggestion (not applied).*
3. **NAC** (already in the stack) and protein with cysteine: support glutathione (GCLM –588T, GPX1 Leu198). *Suggestion (not applied).*
4. **Selenium by feeding** (1–2 chestnuts per day or fish and eggs): cofactor of GPX1 and GPX4; do not add selenium supplements without measuring. *Suggestion (not applied).*
5. **Omega-3 3.2–4 g/day of EPA+DHA**, with olive oil and polyphenols from the feed: protect HDL and PON1 of low expression. *Suggestion (not applied).*
6. **Avoid organophosphates** (domestic insecticides, well-washed fruits and vegetables): Low expression PON1. *Suggestion (not applied).*
7. **Regular exercise** (active FOXO3, NRF2, SOD2 and eNOS) and vegetal nitrates (beet, rocket, spinach) for nitric oxide (NOS3 –786C). *Suggestion (not applied).*
8. **Antioxidants in megadoses** (vitamin E, vitamin C at high doses) are not recommended: with these genes, the strategy is to activate endogenous defenses, not replace them. *Suggestion (not applied).*
9. **Zero alcohol** (It is already the rule). *Suggestion (not applied).*

## 9. Limitations

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- Small and added effects; do not measure real oxidative stress.
- **GSTM1**: The rs366631 marker suggests deletion, but does not measure the number of copies. **GSTT1** is not testable on the chip.
- **Unread**: SOD3 Arg213Gly (rs1799895; out of the chip and without imputation with  $DR2 \geq 0.8$ ; a previous commercial reading gave C/C, without the rare allele) and functional repetitions (NOS3 VNTR, HMOX1 (GT)<sub>n</sub>, GCLC GAG, GPX1 alanines).
- The NRF2 –651 (rs6706649) has called the doubtful chip.
- Seven SNPs come from imputation ( $DR2$  0.94–1.00).
- Conclusions only from genes; do not include blood analysis.

## 10. Methods and sources

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- **Genes and SNPs:** 31 SNPs chosen from the previous analysis of oxidative stress (64 lines), maintaining functional variants; MTHFR and COMT are in the report Methylation, Homocysteine and Hill.
- **Genotypes:** Sérgio F chip (GRCh38, strand +), reread for this report. Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with  $DR2 \geq 0.8$ . No genetic data came out of the analysis computer.
- **Guidance:** Alleles of effect on strand +; in the genes of strand – (SOD2, MPO, PON1, CYBA, NQO1) the nomenclature of the gene was converted. 2 palindromic SNPs assumed in the strand +.
- **Polygenic scores:** PGS Catalog, calculated with the imputed genome; percentiles in the reference population.
- **Comparison with previous analysis (02-10-2026):** The 23 chip genotypes are equal; the 7 imputed (NRF2 –617, PON1 ×2, HMOX1, NOS3 Asp298, MPO, APOE) confirm the previous inference by close neighbors; the SOD3 passed to "unread".
- **Context Literature:** Willcox 2008 and Flachsbart 2009 (FOXO3), Marzec 2007 (NRF2), Bastaki 2006 (SOD2/GPX1), Huang 2009 (GSTM1 marker).

## 11. Annex: all variants analysed

| Topic                                  | Gene           | SNP        | Genotype                  | Source                                         | Effect allele | Copies | Chain    | Interpretation                                            |
|----------------------------------------|----------------|------------|---------------------------|------------------------------------------------|---------------|--------|----------|-----------------------------------------------------------|
| Primary antioxidant enzymes            | SOD2 (MnSOD)   | rs4880     | G/G                       | chip                                           | G             | 2      | strand + | Ala/Ala: MnSOD more active (depends on GPX1 and catalase) |
| Primary antioxidant enzymes            | SOD3 (EC-SOD)  | rs1799895  | SNP not found in the data | SNP not found in the data (DR2<0.8 imputation) | G             | —      | —        | SNP not found in the data                                 |
| Primary antioxidant enzymes            | CAT (catalase) | rs1001179  | C/C                       | chip                                           | T             | 0      | strand + | C/C: common genotype, typical activity                    |
| Primary antioxidant enzymes            | GPX1           | rs1050450  | G/A                       | chip                                           | A             | 1      | strand + | 1 copy of risk allele                                     |
| Primary antioxidant enzymes            | GPX4           | rs713041   | T/C                       | chip                                           | T             | 1      | strand + | 1 copy of effect allele                                   |
| Glutathione: synthesis and recycling   | GCLC           | rs17883901 | G/G                       | chip                                           | A             | 0      | strand + | without the risk allele                                   |
| Glutathione: synthesis and recycling   | GCLM           | rs41303970 | G/A                       | chip                                           | A             | 1      | strand + | 1 copy of risk allele                                     |
| Master regulators (NRF2 and FOXO3)     | NFE2L2 (NRF2)  | rs6721961  | G/G                       | imputation DR2=1.00                            | T             | 0      | strand + | without the risk allele                                   |
| Master regulators (NRF2 and FOXO3)     | NFE2L2 (NRF2)  | rs35652124 | T/C                       | chip                                           | C             | 1      | strand + | 1 copy of risk allele                                     |
| Master regulators (NRF2 and FOXO3)     | NFE2L2 (NRF2)  | rs6706649  | T/T                       | chip                                           | T             | 2      | strand + | 2 risk allele copies                                      |
| Master regulators (NRF2 and FOXO3)     | FOXO3          | rs2802292  | G/G                       | chip                                           | G             | 2      | strand + | 2 copies of the protective allele                         |
| Master regulators (NRF2 and FOXO3)     | FOXO3          | rs2764264  | C/C                       | chip                                           | C             | 2      | strand + | 2 copies of the protective allele                         |
| Master regulators (NRF2 and FOXO3)     | FOXO3          | rs13217795 | C/C                       | chip                                           | C             | 2      | strand + | 2 copies of the protective allele                         |
| Master regulators (NRF2 and FOXO3)     | FOXO3          | rs2802288  | A/A                       | chip                                           | A             | 2      | strand + | 2 copies of the protective allele                         |
| Master regulators (NRF2 and FOXO3)     | FOXO3          | rs12212067 | T/G                       | chip                                           | G             | 1      | strand + | 1 copy of the protective allele                           |
| II phase detoxification (NQO1 and GST) | NQO1           | rs1800566  | G/G                       | chip                                           | A             | 0      | strand + | without the risk allele                                   |
| II phase detoxification (NQO1 and GST) | NQO1           | rs1131341  | G/G                       | chip                                           | A             | 0      | strand + | without the risk allele                                   |
| II phase detoxification (NQO1 and GST) | GSTP1          | rs1695     | A/G                       | chip                                           | G             | 1      | strand + | 1 copy of effect allele                                   |

| Topic                                  | Gene           | SNP       | Genotype | Source                 | Effect allele | Copies | Chain                                | Interpretation                                                                    |
|----------------------------------------|----------------|-----------|----------|------------------------|---------------|--------|--------------------------------------|-----------------------------------------------------------------------------------|
| II phase detoxification (NQO1 and GST) | GSTP1          | rs1138272 | C/C      | chip                   | T             | 0      | strand +                             | without the risk allele                                                           |
| II phase detoxification (NQO1 and GST) | GSTA1          | rs3957357 | A/A      | chip                   | A             | 2      | strand +                             | 2 risk allele copies                                                              |
| II phase detoxification (NQO1 and GST) | GSTM1          | rs366631  | A/A      | chip                   | A             | 2      | strand +                             | Probable null GSTM1 (marker)                                                      |
| Paraoxonase and hemeoxygenase          | PON1           | rs662     | T/T      | chip                   | C             | 0      | strand +                             | Gln/Gln: better protection of lipoproteins; less defense against organophosphates |
| Paraoxonase and hemeoxygenase          | PON1           | rs854560  | T/T      | imputation<br>DR2=1.00 | T             | 2      | strand +<br>(palindromic; assumed +) | 2 risk allele copies                                                              |
| Paraoxonase and hemeoxygenase          | PON1           | rs705379  | A/A      | imputation<br>DR2=0.97 | A             | 2      | strand +                             | 2 risk allele copies                                                              |
| Paraoxonase and hemeoxygenase          | HMOX1          | rs2071746 | A/A      | imputation<br>DR2=0.94 | T             | 0      | strand +<br>(palindromic; assumed +) | without the risk allele                                                           |
| Oxidant sources and nitric oxide       | NOS3 (eNOS)    | rs2070744 | C/T      | chip                   | C             | 1      | strand +                             | 1 copy of risk allele                                                             |
| Oxidant sources and nitric oxide       | NOS3 (eNOS)    | rs1799983 | T/G      | imputation<br>DR2=1.00 | T             | 1      | strand +                             | 1 copy of risk allele                                                             |
| Oxidant sources and nitric oxide       | NOS3 (eNOS)    | rs3918226 | C/C      | chip                   | T             | 0      | strand +                             | without the risk allele                                                           |
| Oxidant sources and nitric oxide       | MPO            | rs2333227 | C/C      | imputation<br>DR2=1.00 | C             | 2      | strand +                             | 2 risk allele copies                                                              |
| Oxidant sources and nitric oxide       | CYBA (p22phox) | rs4673    | G/G      | chip                   | A             | 0      | strand +                             | without the protective allele (common genotype)                                   |
| Oxidant sources and nitric oxide       | APOE           | rs429358  | T/T      | imputation<br>DR2=0.99 | C             | 0      | strand +                             | without the risk allele                                                           |